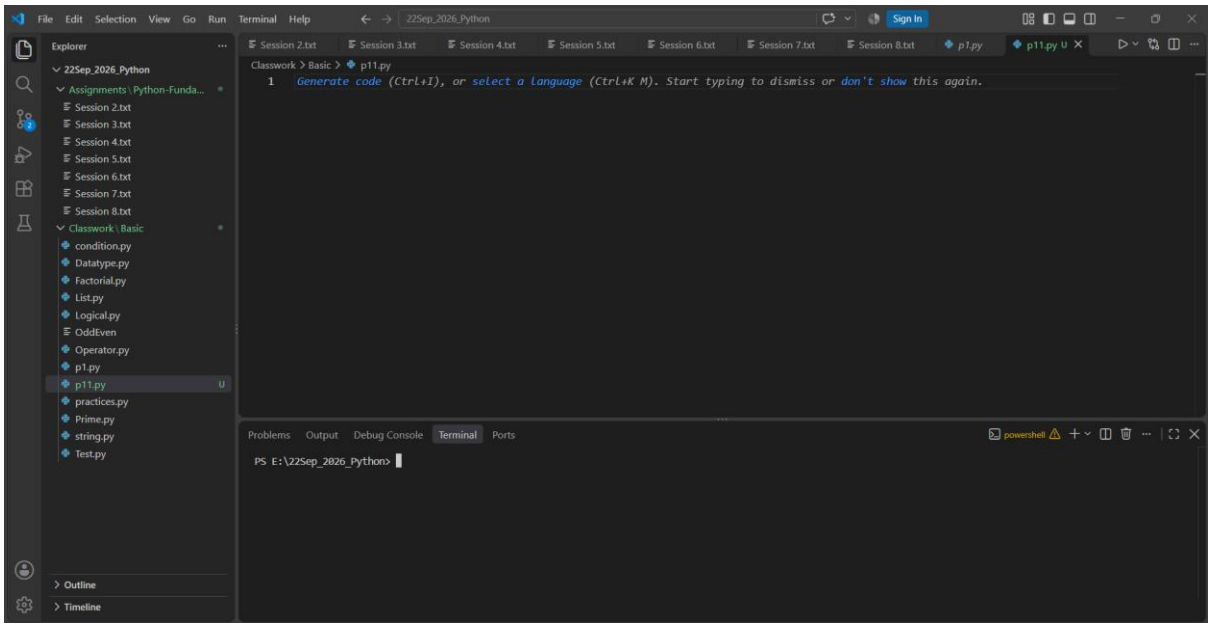
Session 1: Introduction to Python what is Python? Why Python? Use cases installing Python & VS Code Running first Python script Python code execution model.

TASKS:

Question-1: Install Python (latest version) and VS Code on your computer, then take a screenshot of both the Python installation confirmation and the VS Code welcome screen.

Ans:





Question-2: Create a new Python file named hello_instagram.py in VS Code and write code to print 'Hello, Instagram World!' to the terminal.

Ans:

print(“Hello, Instagram world!”)

Question-3: Write a Python script called app_use_cases.py that prints three real-life apps you use daily and, for each, prints one way Python could be used in their backend or data analysis.

Ans:

```
print("WhatsApp")
print("Python handles real-time messaging.")

print("YouTube")
print("Python recommends videos.")

print("Instagram")
print("Python manages user activity.")
```

Question-4: Modify your hello_instagram.py script to print the Python version currently running your script.

   Hint: Use the sys module and sys.version.

Ans:

```
import sys
print(sys.version)
```

Question-5: Explain the difference between how Python executes code versus how C or Java executes code, using your own words and a simple example for each.

Ans:

Difference between Python, C, and Java

The primary difference is how they execute code. Python uses an interpreter, C compiles code into machine language, and Java compiles code into bytecode that runs on the JVM.

1. Python: Interpreter

Python executes code using an interpreter while the program runs.

Example:

```
print("Hello World")
```

2. C: Compiler

C code is compiled into machine code before execution. If a compilation error occurs, the program cannot be built successfully.

Example:

```
#include <stdio.h>

int main() {
   printf("Hello World");
   return 0;
}
```

3. Java: Bytecode and JVM

Java code is compiled into bytecode, which runs on the Java Virtual Machine (JVM). This allows Java programs to run on different operating systems that have a compatible JVM.

Example:

```
class Main {
   public static void main(String[] args) {
      System.out.println("Hello World");
   }
}
```

Difference between Python, C, and Java

Python	**C**	**Java**
Python uses an interpreter and Python runtime.	C uses a compiler to generate machine code.	Java compiles code into bytecode.
Python code is generally easy to read and write.	C provides low-level memory control.	Java code runs on the JVM.
Python is commonly used for AI, automation, and data analysis.	C is commonly used for operating systems and embedded systems.	Java is commonly used for enterprise and Android applications.
Example: print("Hello")	Example: printf("Hello");	Example: System.out.println("Hello");